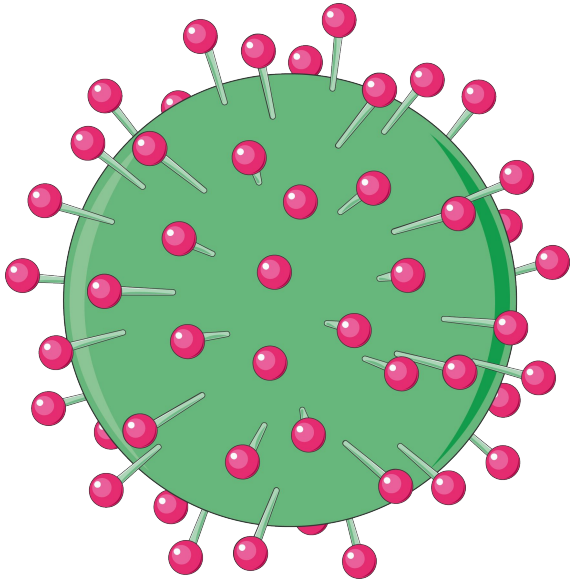
HEPATITIS B VIRAL CAPSID PROTEIN

HBcAg



Alessandra Bonilla, Marc J. Torres & Oscar Contreras

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1. INTRODUCTION

- Our project consists on the functional, sequential and structural analysis of Hepatitis Virus Capsid using bioinformatics tools
- Our choice → Hepatitis B
  - Ability to form a **superstructure of capsids**
  - Capsid protein: HBcAg

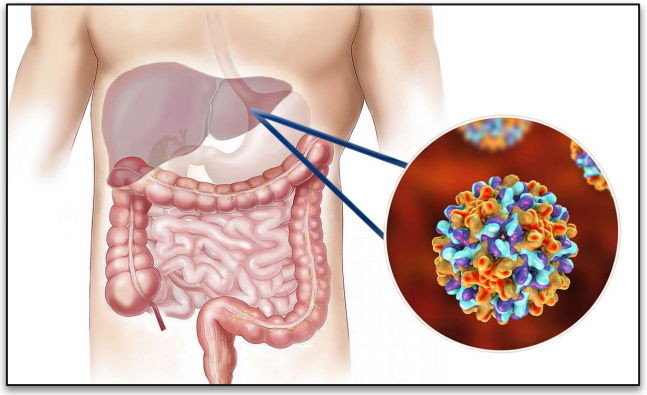


Figure 1: Hepatitis is a liver infectious disease caused by Hepatitis B virus, Studika.

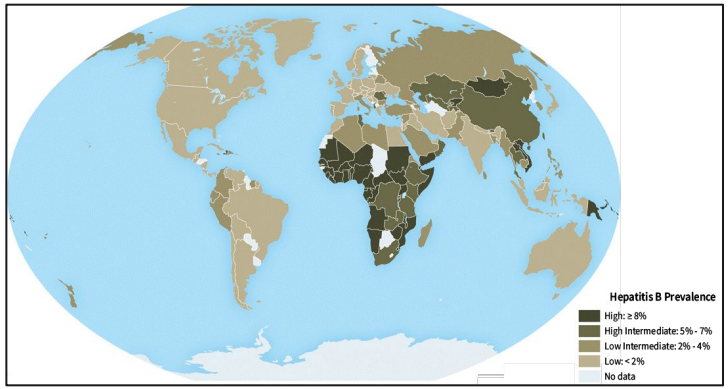


Figure 2 : Countries most affected by hepatitis B

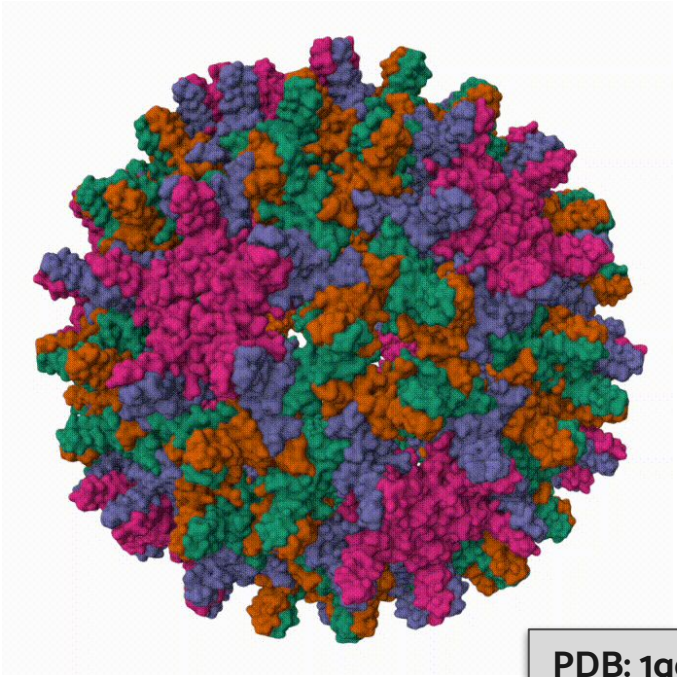
2. GENERAL ASPECTS

HEPATITIS B

- Inflammatory liver disease
- Hepatitis B virus (HBV) → Hepadnaviridae family

HBcAg

- Unique ability to form a **superstructure of capsids**
- Icosahedral capsid
- Function of hepatitis B core protein is multifaceted in the viral life cycle
  - Self-assemblies
    - Protective shell and transport



PDB: 1qgt

HOW DOES IT CARRY OUT ITS FUNCTION?

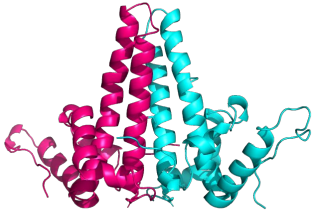
CAPSID ASSEMBLY

01

In order to be able to do the reverse transcription, the capsid formation is essential

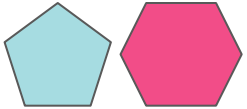
02

Arrangement of the dimers



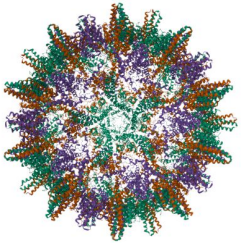
03

Formation of of pentagonal and quasi-hexagonal structure



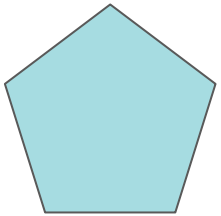
04

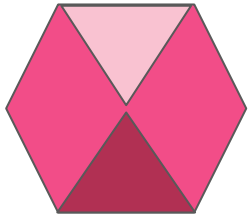
Assembly of the superstructure by burial of hydrophobic surface areas



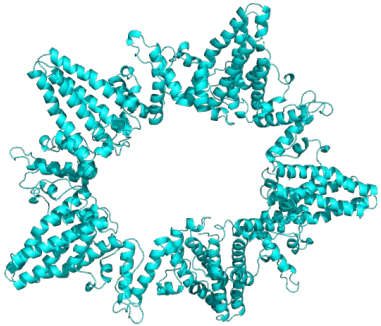
UNDERSTANDING ITS STRUCTURE

- Icosahedral symmetry
- 12 vertices, 20 faces and 30 edges
- T = 4 structure (sometimes T = 3)
- 120 monomers (60 asymmetric unit)
- Arrangement of **A**, **B**, **C**, and **D** monomers → pentagonal and quasi-hexagonal vertices

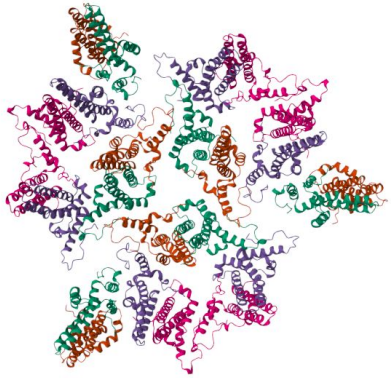




Icosahedral pentamer



Icosahedral quasi- hexamer



N-terminal α1 α2

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O71303|Lagothrix_lagotricha        MDIDPYKEFGATVELLSFLPADFFPSVRDLLDTASALYREALESSDHCSP
P0C6J1|Otospermophilus_beechey     MDIDPYKEFGSSYQLLNFLPLDFFPDLNALVDTAAALYEEELTGREHCSP
P69709|Marmota_monax               MDIDPYKEFGSSYQLLNFLPLDFFPDLNALVDTATALYEEELTGREHCSP
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                                            α3                      α4
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O71303|Lagothrix_lagotricha        HHTALRQTVLCWGELMSLASWVCTNLEDPAARELVVSYVNDNMGLKVRQL
P0C6J1|Otospermophilus_beechey     HHTAIRQALVCWEELTRLITWMSEN-TTEEVRRIIVDHVNNTWGLKVRQT
P69709|Marmota_monax               HHTAIRQALVCWDELTKLIAWMSSNITSEQVRTIIVNHVNDTWGLKVRQS
                                   :***:**:::** **  * :*:. *      *  :*.:** . ***:**
                                                     α5
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P12901|Pan_troglodytes             LWFHISCLTFGRETVLEYLVSFGVWIRTPPAYRPPNAPILSTLPETTVVR
O71303|Lagothrix_lagotricha        LWFHISCLTFGRETVLEYLVSFWVWIRTPPAYRPPNAPILSTLPETTVVR
P0C6J1|Otospermophilus_beechey     LWFHLSCLTFGQHTVQEFLVSFGVWIRTPAPYRPPNAPILSTLPEHTVIR
P69709|Marmota_monax               LWFHLSCLTFGQHTVQEFLVSFGVWIRTPAPYRPPNAPILSTLPEHTVIR
                                   ****:******:.** *:**** ******..************** **:*

P03149|Homo_sapiens                RRD---RGRSPRRRTPSPRRRRSQSPRRRRSQSRESQC
P12901|Pan_troglodytes             RR-----GRSPRRRTPSPRRRRSQSPRRRRSQSPASQC
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P69709|Marmota_monax               RRGGARASRSPRRRTPSPRRRRSQSPRRRRSQSPSANC
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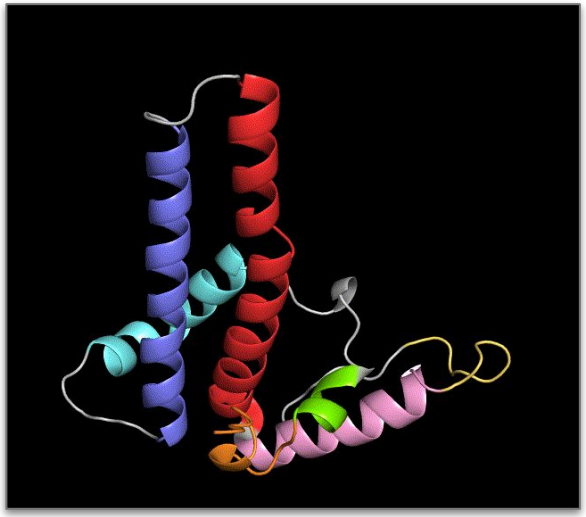

3. CONSERVED REGIONS

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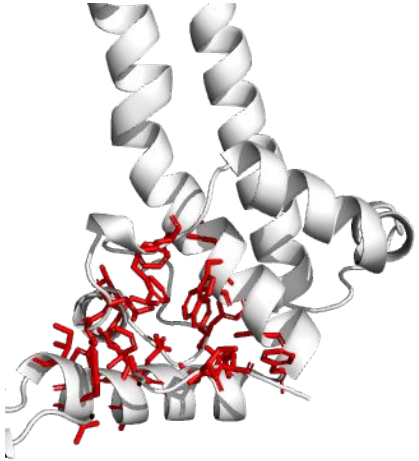
P12901|Pan_troglodytes             MDIDPYKEFGATVELLSFLPSDFFPSVRDLLDTASALYREALESPEHCSP
O71303|Lagothrix_lagotricha        MDIDPYKEFGATVELLSFLPADFFPSVRDLLDTASALYREALESSDHCSP
P0C6J1|Otospermophilus_beechey     MDIDPYKEFGSSYQLLNFLPLDFFPDLNALVDTAAALYEEELTGREHCSP
P69709|Marmota_monax               MDIDPYKEFGSSYQLLNFLPLDFFPDLNALVDTATALYEEELTGREHCSP
                                   **********:: :**.*** ****.:. *:***:***.* * . :****

P03149|Homo_sapiens                HHTALRQAILCWGELMTLATWVGNNLQDPASRDLVVNYVNTNMGLKIRQL
P12901|Pan_troglodytes             NHTALRQAILCWGELMTLASWVGNNLEDPASREQVVNYVNTNMGLKIRQL
O71303|Lagothrix_lagotricha        HHTALRQTVLCWGELMSLASWVGTNLEDPAARELVVSYVNDNMGLKVRQL
P0C6J1|Otospermophilus_beechey     HHTAIRQALVCWEELTRLITWMSEN-TTEEVRRIIVDHVNNTWGLKVRQT
P69709|Marmota_monax               HHTAIRQALVCWDELTKLIAWMSSNITSEQVRTIIVNHVNDTWGLKVRQS
                                   :***:**:::** **  * :*:. *      *  :*.:** . ***:**

P03149|Homo_sapiens                LWFHISCLTFGRETVLEYLVSFGVWIRTPPAYRPPNAPILSTLPETTVVR
P12901|Pan_troglodytes             LWFHISCLTFGRETVLEYLVSFGVWIRTPPAYRPPNAPILSTLPETTVVR
O71303|Lagothrix_lagotricha        LWFHISCLTFGRETVLEYLVSFWVWIRTPPAYRPPNAPILSTLPETTVVR
P0C6J1|Otospermophilus_beechey     LWFHLSCLTFGQHTVQEFLVSFGVWIRTPAPYRPPNAPILSTLPEHTVIR
P69709|Marmota_monax               LWFHLSCLTFGQHTVQEFLVSFGVWIRTPAPYRPPNAPILSTLPEHTVIR
                                   ****:******:.** *:**** ******..************** **:*

P03149|Homo_sapiens                RRD---RGRSPRRRTPSPRRRRSQSPRRRRSQSRESQC
P12901|Pan_troglodytes             RR-----GRSPRRRTPSPRRRRSQSPRRRRSQSPASQC
O71303|Lagothrix_lagotricha        RR------RPSGRRTPSPRRRRSQSPRRRRSQSPASSC
P0C6J1|Otospermophilus_beechey     RRGGSRAARSPRRRTPSPRRRRSQSPRRRRSQSPASNC
P69709|Marmota_monax               RRGGARASRSPRRRTPSPRRRRSQSPRRRRSQSPSANC
                                   **      *.. ********************  :.*
```

- Highly conservation of amino acids forming the **hydrophobic core** of HBcAG, crucial to monomer stability



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```
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O71303|Lagothrix_lagotricha        MDIDPYKEFGATVELLSFLPADFFPSVRDLLDTASALYREALESSDHCSP
P0C6J1|Otospermophilus_beechey     MDIDPYKEFGSSYQLLNFLPLDFFPDLNALVDTAAALYEEELTGREHCSP
P69709|Marmota_monax               MDIDPYKEFGSSYQLLNFLPLDFFPDLNALVDTATALYEEELTGREHCSP
                                   **********:: :**.*** ****.:. *:***:***.* * . :****

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P12901|Pan_troglodytes             NHTALRQAILCWGELMTLASWVGNNLEDPASREQVVNYVNTNMGLKIRQL
O71303|Lagothrix_lagotricha        HHTALRQTVLCWGELMSLASWVGTNLEDPAARELVVSYVNDNMGLKVRQL
P0C6J1|Otospermophilus_beechey     HHTAIRQALVCWEELTRLITWMSEN-TTEEVRRIIVDHVNNTWGLKVRQT
P69709|Marmota_monax               HHTAIRQALVCWDELTKLIAWMSSNITSEQVRTIIVNHVNDTWGLKVRQS
                                   :***:**:::** **  * :*:. *      *  :*.:** . ***:**

P03149|Homo_sapiens                LWFHISCLTFGRETVLEYLVSFGVWIRTPPAYRPPNAPILSTLPETTVVR
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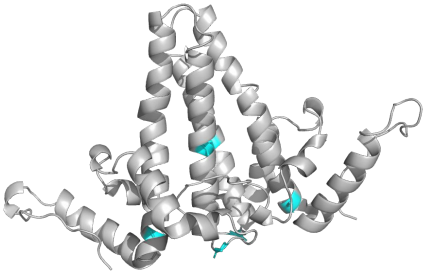
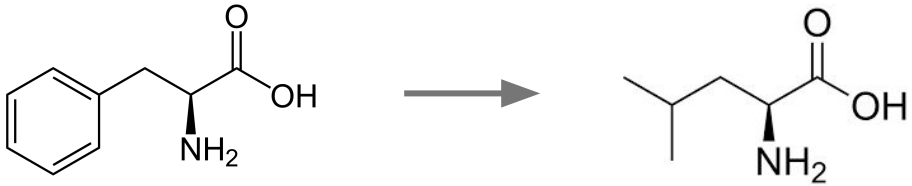
O71303|Lagothrix_lagotricha        LWFHISCLTFGRETVLEYLVSFWVWIRTPPAYRPPNAPILSTLPETTVVR
P0C6J1|Otospermophilus_beechey     LWFHLSCLTFGQHTVQEFLVSFGVWIRTPAPYRPPNAPILSTLPEHTVIR
P69709|Marmota_monax               LWFHLSCLTFGQHTVQEFLVSFGVWIRTPAPYRPPNAPILSTLPEHTVIR
                                   ****:******:.** *:**** ******..************** **:*

P03149|Homo_sapiens                RRD---RGRSPRRRTPSPRRRRSQSPRRRRSQSRESQC
P12901|Pan_troglodytes             RR-----GRSPRRRTPSPRRRRSQSPRRRRSQSPASQC
O71303|Lagothrix_lagotricha        RR------RPSGRRTPSPRRRRSQSPRRRRSQSPASSC
P0C6J1|Otospermophilus_beechey     RRGGSRAARSPRRRTPSPRRRRSQSPRRRRSQSPASNC
P69709|Marmota_monax               RRGGARASRSPRRRTPSPRRRRSQSPRRRRSQSPSANC
                                   **      *.. ********************  :.*
```

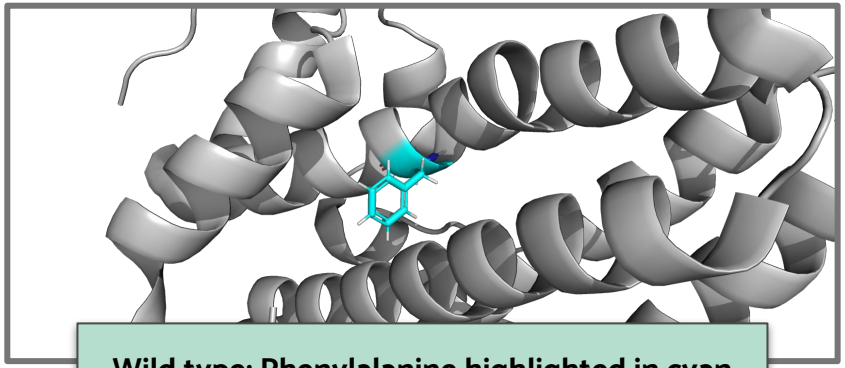
- **Cysteines** conserved across all the genotypes and in different species
- **Location** → bonds between alpha helix hairpins and dimer base
- Formation **disulfide bonds** → stabilization the two alpha helix hairpins of the dimer by forming covalent links



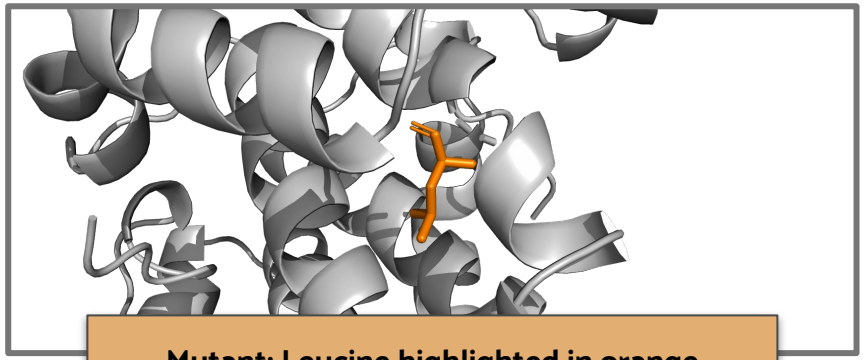
4. EXPLORING A MUTATION



▼ Mutagenesis	97	Enhances capsid assembly. 1 Publication
Sequence: F → L		



Wild type: Phenylalanine highlighted in cyan



Mutant: Leucine highlighted in orange

TABLE 1.

Thermodynamic parameters describing assembly of Cp149-wt and -F971L[a]

Protein	ΔG (kcal/mol) at: 25°	ΔG (kcal/mol) at: 37°	ΔH (kcal/mol)	$-T\Delta S$ (kcal/mol) at 37°	$1/K_{contact}$ (mM) at: 25°	$1/K_{contact}$ (mM) at: 37°	$K_{Dapparent}$ (μM) at 37°
Cp149-wt	−3.0	−3.27	+2.0	−5.27	6.3	5.0	8.3
Cp149-F97L	−3.0	−3.6	+7.4	−11.0	5.4	3.0	2.0

Open in a separate window

[a]Values are per contact. Temperatures are in degrees Celsius.

Figure 2: Ceres, P., Stray, S. J., & Zlotnick, A. (2004). Hepatitis B virus capsid assembly is enhanced by naturally occurring mutation F97L. Journal of virology, 78(17), 9538–9543. https://doi.org/10.1128/JVI.78.17.9538-9543.2004

Mutant has less Gibbs energy so, it is more stable, from a thermodynamic point of view.

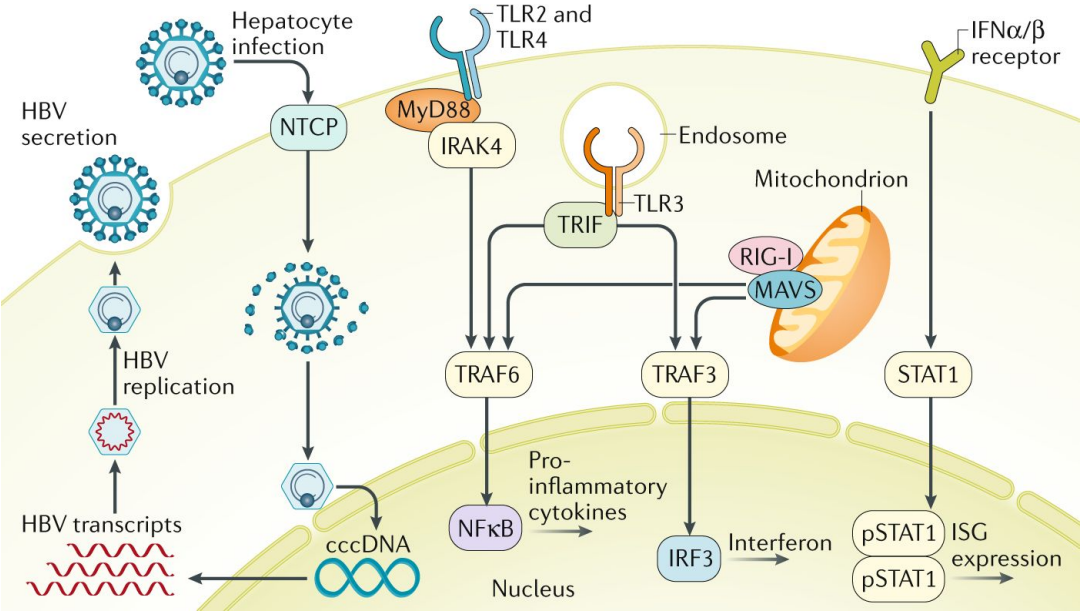
$$\Delta G = \Delta H - T\Delta S$$

$$\downarrow$$

$$\uparrow$$

5. INTERACTIONS

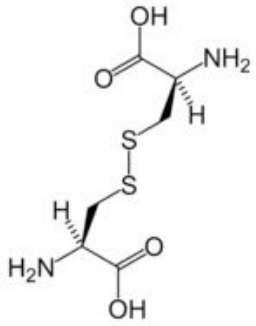
- **Host cell proteins and molecules**
- Viral proteins
- cccDNA
- Heparan sulfate proteoglycans (HSPGs)
- Sodium taurocholate co-transporting polypeptide (NTCP)
- **The most basic interaction with HBcAg protein (itself)**

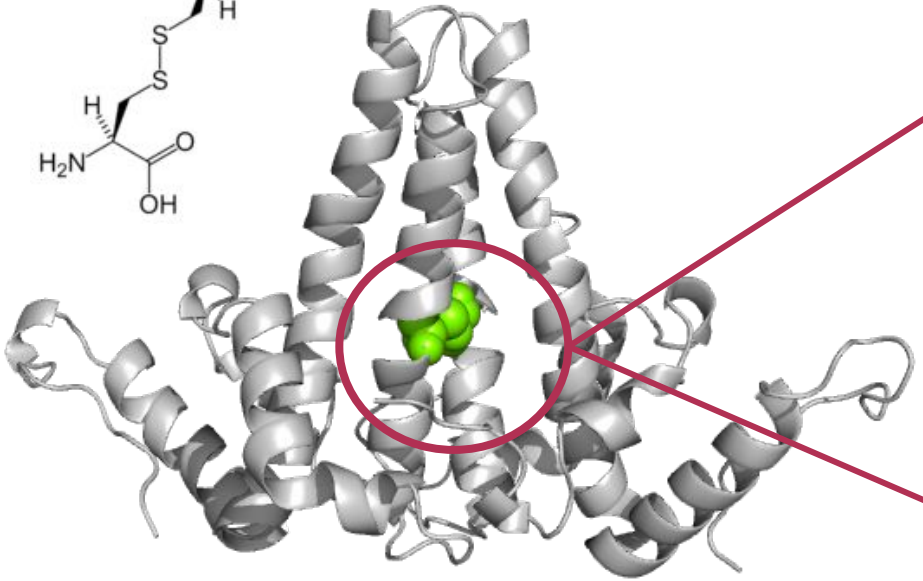


Host–pathogen interactions in chronic HBV infection

INTERACTION WITHIN THE DIMER

CYSTEINE AT POSITION 61



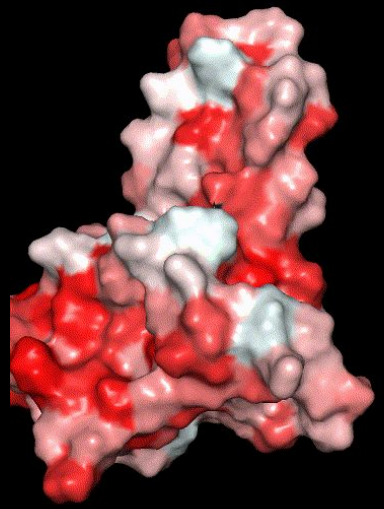


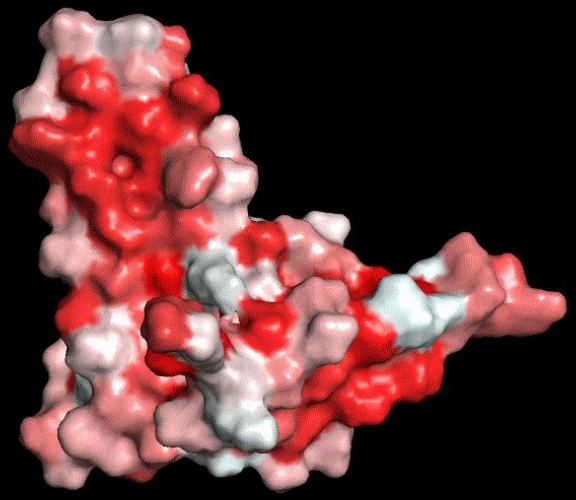
S.A Wynne, R.A Crowther, A.G.W Leslie, The Crystal Structure of the Human Hepatitis B Virus Capsid, Molecular Cell, Volume 3, Issue 6, 1999, Pages 771-780, ISSN 1097-2765, https://doi.org/10.1016/S1097-2765(01)80009-5.

INTERACTION BETWEEN MONOMERS

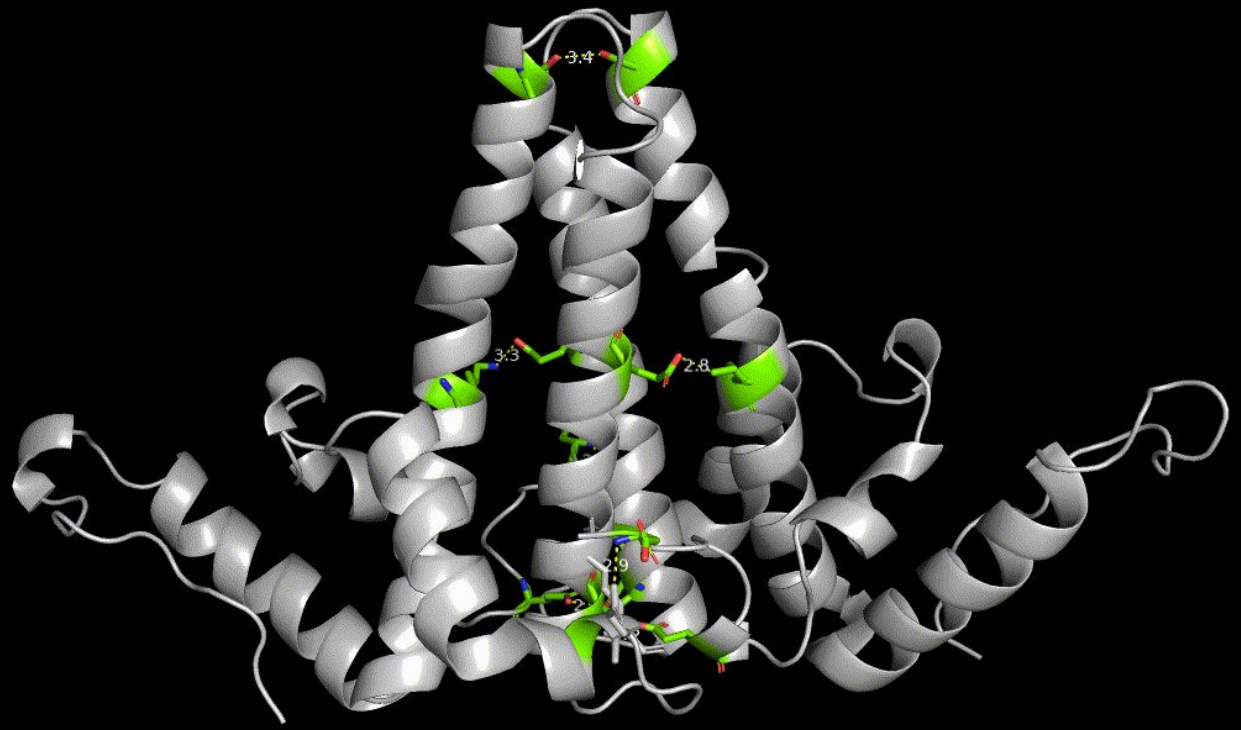
Surfaces of two monomers in which amino acids are colored based on hydrophobicity scale .

- More hydrophobic
- Quite hydrophobic
- No hydrophobic

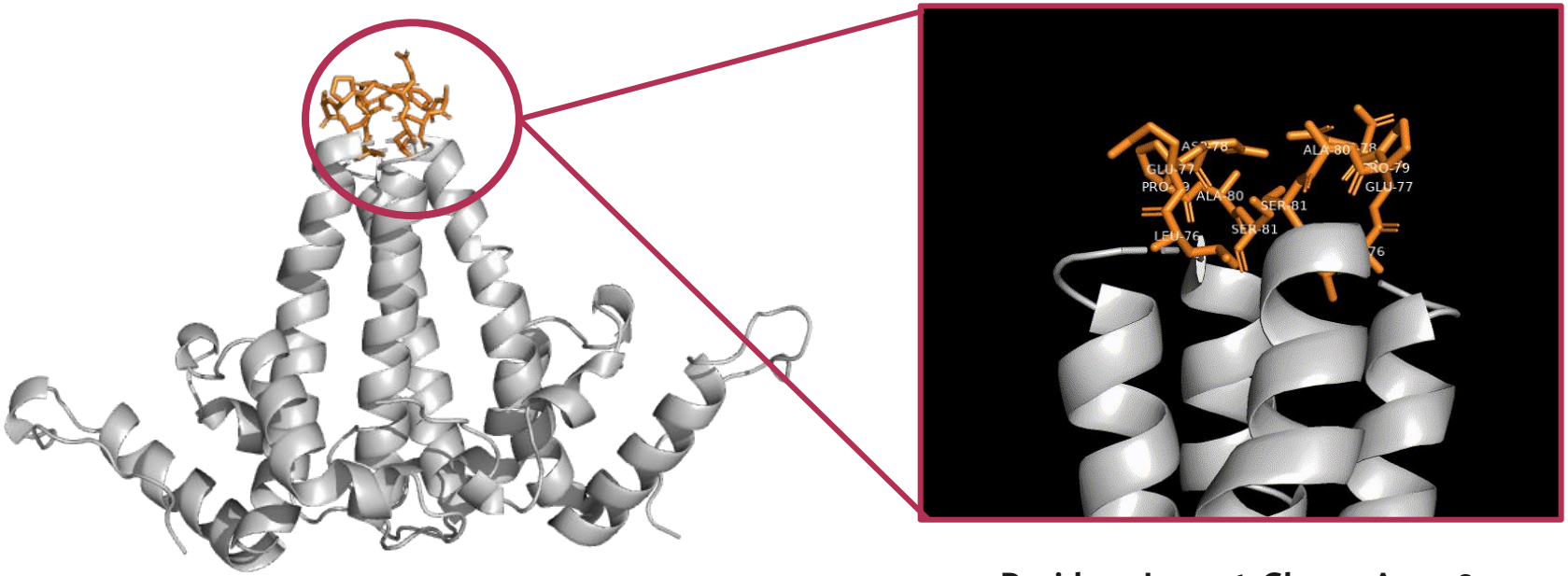




INTERACTION BETWEEN MONOMERS



HBcAg region that binds an antibody



Residues Leu-76, Glu-77, Asp-78, Pro-79, Ala-80, and Ser-81

6. SUPERIMPOSITIONS

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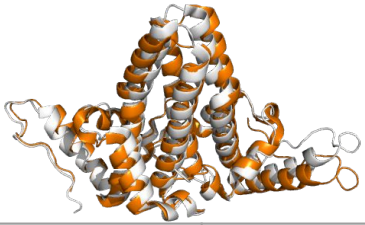
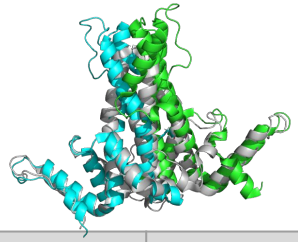
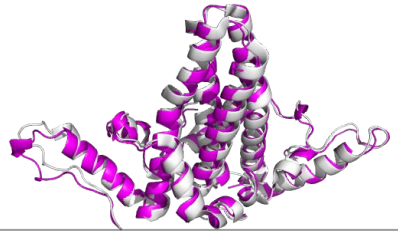
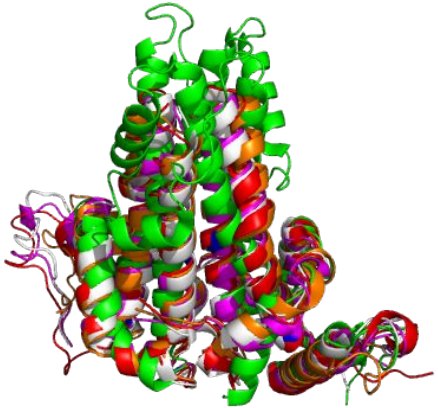
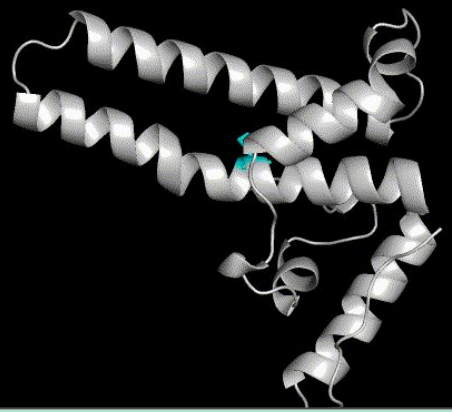
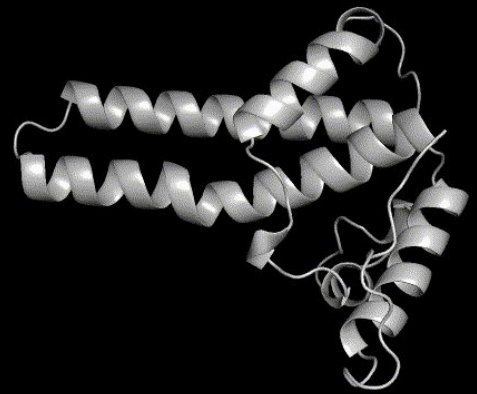
			
PDB IDs	RMSD	PDB IDs	RMSD
1QGT, 6UI6	1.300	2G33, 6UI6	1.587
			



6. HOMOLOGOUS MODELLING

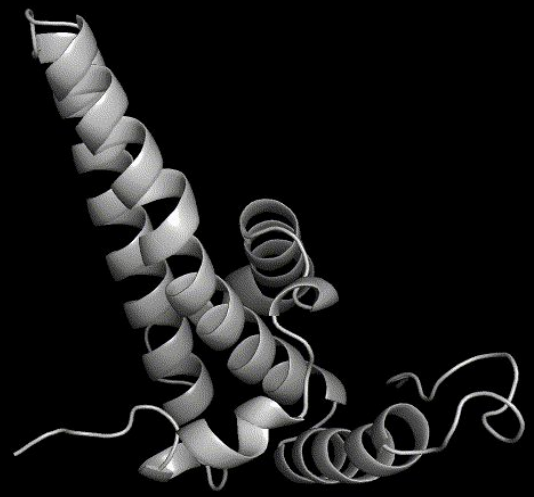


Modeller (Wild type): Isoleucine

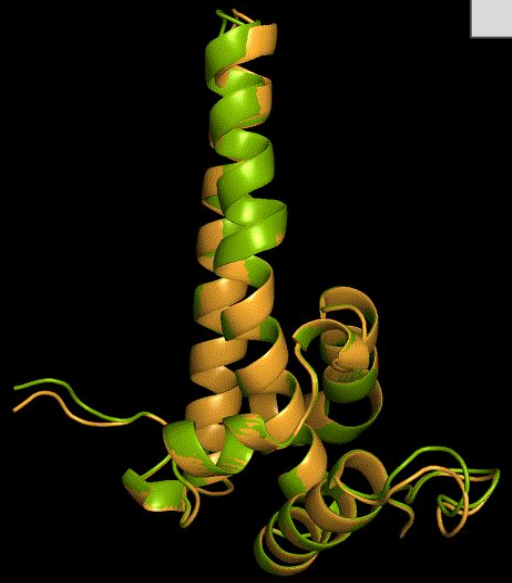


Modeller (Wild type): Phenylalanine

6. HOMOLOGOUS MODELLING



Modeller (Mutant): Leucine

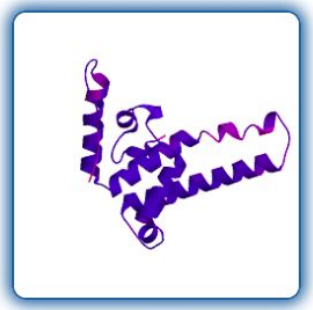


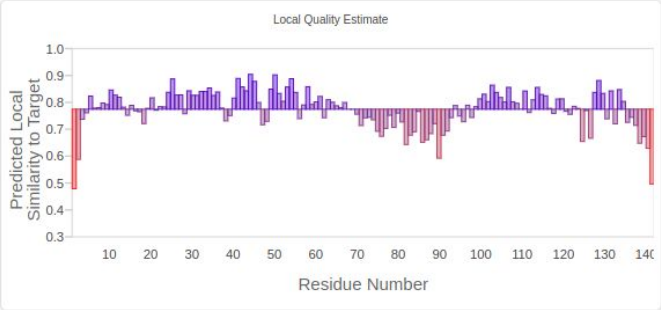
Superimposing mutant and wild type

QMEAN: MUTANT (LEUCINE)

Quality for P03149.B99990002.pdb

Downloads



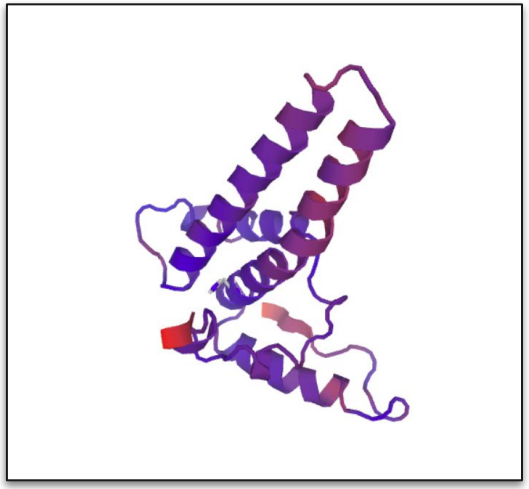


QMEANDisCo Global: 0.77 ± 0.07

Sequence colored by local quality:

Local quality score
0.0 0.5 1.0

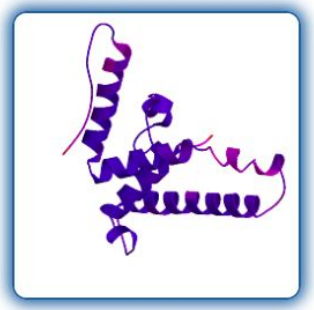
A: MDIDPYKEFGATVELLSFLPSDFFPSVRDLLDTASALYREALESPEHCSPHHTALRQAILCWGELMTLATWVGNN 75
A: LQDPASRDLVVNYVNTNMGLKIRQLLWFHISCLTFGRETVLEYLVSFGVWIRTPPAYRPPNAPILST 142

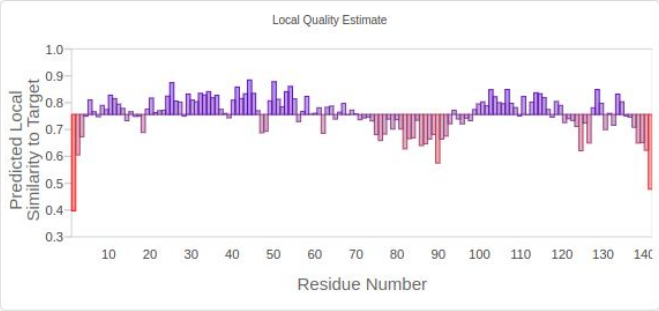


QMEAN: WILD TYPE (PHENYLALANINE)

Quality for P03149.B99990002.pdb

Downloads





QMEANDisCo Global: 0.76 ± 0.07

Sequence colored by local quality:

Local quality score
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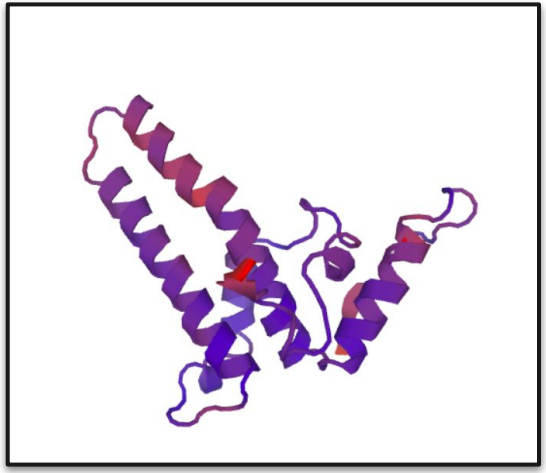

CONCLUSIONS

- The capsid has an icosahedron shape formed by the repetition of dimers.
- The HBcAg is very unique and has the property of self assembly.
- We have identified conserved regions in the hepatitis capsid protein such as the hydrophobic core and the cysteines.
- The hepatitis B virus capsid is highly conserved along different species.